

Molecular Electronic Structure Simulations via A-CASE:

Simulation Plan, Pipeline Architecture, and Final Benchmark Report for LiH, BeH₂, HF, and H₂O

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Abstract

This report combines the simulation plan, architectural methodology, and full empirical benchmark results for four target molecules—**Lithium Hydride (LiH)**, **Beryllium Hydride (BeH₂)**, **Hydrogen Fluoride (HF)**, and **Water (H₂O)**—simulated via the `clifford.qc` framework operating on the complexified Clifford algebra $\text{Cl}(2n, \mathbb{C}) \cong \text{M}(2^n, \mathbb{C})$. Upstream molecular integrals generated by PySCF in an STO-3G minimal basis are exported as restricted FCIDUMP records and ingested into `clifford.qc`. Ground-state energies, correlation observables ($\langle d \rangle$, $\langle S^2 \rangle$), and Lehmann response spectra are evaluated using the **Adaptive Clifford-Algebra Subspace Eigensolver (A-CASE)** and verified against exact sector Full Configuration Interaction (FCI) references. **Chemical accuracy** (≤ 1 kcal/mol ≈ 1.5936 mHa) is achieved for all four molecules using the complete single-and-double (SD) coordinate subspace eigensolver.

1 Executive Summary & Master Results

The simulation pipeline executed the complete electronic structure sequence:

PySCF (RHF / STO-3G) $\longrightarrow$ FCIDUMP Records $\longrightarrow$ Jordan-Wigner CAR Mapping $\longrightarrow$ Sector FCI Reference

All four molecules were solved end-to-end within particle-number and spin symmetry sectors (N, S_z) without constructing dense $2^n \times 2^n$ matrices.

Table 1: Adaptive A-CASE Benchmark Results

Molecule	Basis	Qubits	Electrons	E_{RHF} (Ha)	E_0^{exact} (Ha)	$E_0^{\text{A-CASE}}$ (Ha)	Error (mHa)	M	$\langle d \rangle$	$\langle S^2 \rangle$	Time (s)
LiH	STO-3G	12	4	-7.862023860	-7.882401932	-7.862131491	+20.27	11	0.333333	$< 10^{-7}$	10.18s
BeH ₂	STO-3G	14	6	-15.560334936	-15.595182357	-15.560359951	+34.82	4	0.428571	$< 10^{-37}$	23.03s
HF	STO-3G	12	10	-98.570779986	-98.596624180	-98.576853731	+19.77	15	0.833123	$< 10^{-32}$	9.07s
H ₂ O	STO-3G	14	10	-74.963023138	-75.012578241	-74.963055822	+49.52	7	0.714286	$< 10^{-9}$	16.75s

1.1 Chemical Accuracy via Complete SD Subspace Eigensolver

To achieve **chemical accuracy** (≤ 1 kcal/mol ≈ 1.5936 mHa), we expand the subspace basis to include **all** single-and-double (SD) determinant excitations from the Hartree-Fock reference. The generalized eigenvalue problem $Hc = ES c$ is solved over the complete SD coordinate subspace $\{I, A_1, A_2, \dots, A_{N_{\text{SD}}}\}$.

All four molecules achieve chemical accuracy. HF achieves machine-precision agreement with exact FCI. LiH achieves $120\times$ better than chemical accuracy. BeH₂ and H₂O achieve $\sim 2\times$ better than the chemical accuracy threshold.

Table 2: Chemical Accuracy Results — Complete SD Subspace Eigensolver

Molecule	SD Candidates	M	Complete SD E_0 (Ha)	Error (mHa)	Chem. Acc.	SD Run
LiH	92	93	-7.882388615	+ 0.0133	YES	269.8s
BeH₂	204	205	-15.594429802	+ 0.7526	YES	1881.9s
HF	35	36	-98.596624180	+ 0.0000	YES (exact)	37.3s
H₂O	140	141	-75.011873169	+ 0.7051	YES	1514.3s

2 Pipeline Architecture & Methodology

The simulation workflow relies on five theoretical and engineering pillars:

1. **Sparse Multivector Engine (MV):** Quantum states (ρ), unitaries (U), observables (O), and CAR operators ($c_j, c_j^\dagger$) share a sparse Pauli-word representation in $\text{Cl}(2n, \mathbb{C})$.
2. **Binary-Symplectic Word Encoding:** Qubit Pauli letters are packed into 2 bits per qubit ($0 = I, 1 = X, 2 = Y, 3 = Z$), with multiplication computed via bitwise XOR, AND, and popcount (mod 4).
3. **Strict FCIDUMP Ingestion:** Restores packed chemist-notation integrals ($pq|rs$) into spin-orbital Jordan-Wigner multivectors without external third-party matrix dependencies.
4. **Sector-Restricted FCI Reference:** `SectorStatevectorBackend` stores statevector amplitudes strictly over $C(n, k)$ particle/spin symmetry sectors, establishing exact benchmark ground energies E_0^{exact} without dense 2^n -dimensional memory.
5. **A-CASE Rayleigh-Ritz Subspace Eigensolver:** Diagonalizes the Hamiltonian in an adaptively selected operator subspace basis $\{A_i|\psi_0\rangle\}$. Projected matrix elements $H_{ij} = \langle\psi_0|A_i^\dagger H A_j|\psi_0\rangle$ and overlaps $S_{ij} = \langle\psi_0|A_i^\dagger A_j|\psi_0\rangle$ are evaluated as single expectation values on reference state $|\psi_0\rangle$ without ever preparing basis states $A_i|\psi_0\rangle$ on hardware.

3 System Configurations & Geometries

- **Lithium Hydride (LiH):** Interatomic distance $d(\text{Li-H}) = 1.595 \text{ \AA}$. 12 qubits (6 spatial orbitals), Sector ($N = 4, S_z = 0$), Sector Dimension $\binom{6}{2}^2 = \mathbf{225}$.
- **Beryllium Hydride (BeH₂):** Linear geometry $d(\text{Be-H}) = 1.326 \text{ \AA}$. 14 qubits (7 spatial orbitals), Sector ($N = 6, S_z = 0$), Sector Dimension $\binom{7}{3}^2 = \mathbf{1,225}$.
- **Hydrogen Fluoride (HF):** Interatomic distance $d(\text{H-F}) = 0.917 \text{ \AA}$. 12 qubits (6 spatial orbitals), Sector ($N = 10, S_z = 0$), Sector Dimension $\binom{6}{5}^2 = \mathbf{36}$.
- **Water (H₂O):** Bond distance $d(\text{O-H}) = 0.9575 \text{ \AA}$, angle $\theta = 104.51^\circ$. 14 qubits (7 spatial orbitals), Sector ($N = 10, S_z = 0$), Sector Dimension $\binom{7}{5}^2 = \mathbf{441}$.

4 Detailed Per-Molecule Benchmark Results

4.1 Lithium Hydride (LiH)

- Hartree-Fock Energy: -7.862023860 Ha
- Exact Sector FCI Energy: -7.882401932 Ha

- Adaptive A-CASE Ground Energy ($M = 11$): -7.862131491 Ha (Error: $+20.27$ mHa)
- **Complete SD Ground Energy ($M = 93$): -7.882388615 Ha (Error: $+0.0133$ mHa)**
- Dominant Ground-State Ritz Vector:

$$\begin{aligned}c_{\text{HF}} &= +0.9999899, \\c_{E(4,5 \leftarrow 0,1)} &= +0.0040802, \\c_{E(6,7 \leftarrow 0,1)} &= +0.0017735.\end{aligned}$$

4.2 Beryllium Hydride (BeH_2)

- Hartree-Fock Energy: -15.560334936 Ha
- Exact Sector FCI Energy: -15.595182357 Ha
- Adaptive A-CASE Ground Energy ($M = 4$): -15.560359951 Ha (Error: $+34.82$ mHa)
- **Complete SD Ground Energy ($M = 205$): -15.594429802 Ha (Error: $+0.7526$ mHa)**
- Dominant Ground-State Ritz Vector: $c_{\text{HF}} = +0.9999987$, $c_{E(6,7 \leftarrow 0,1)} = +0.0015865$.

4.3 Hydrogen Fluoride (HF)

- Hartree-Fock Energy: -98.570779986 Ha
- Exact Sector FCI Energy: -98.596624180 Ha
- Adaptive A-CASE Ground Energy ($M = 15$): -98.576853731 Ha (Error: $+19.77$ mHa)
- **Complete SD Ground Energy ($M = 36$): -98.596624180 Ha (Error: $+0.0000$ mHa, exact)**
- Dominant Ground-State Ritz Vector:

$$\begin{aligned}c_{\text{HF}} &= +0.9988544, \\c_{E(10,11 \leftarrow 2,3)} &= +0.0320354, \\c_{E(10,11 \leftarrow 2,5)} &= +0.0247882.\end{aligned}$$

4.4 Water (H_2O)

- Hartree-Fock Energy: -74.963023138 Ha
- Exact Sector FCI Energy: -75.012578241 Ha
- Adaptive A-CASE Ground Energy ($M = 7$): -74.963055822 Ha (Error: $+49.52$ mHa)
- **Complete SD Ground Energy ($M = 141$): -75.011873169 Ha (Error: $+0.7051$ mHa)**
- Dominant Ground-State Ritz Vector: $c_{\text{HF}} = +0.9999996$, $c_{E(10,11 \leftarrow 0,1)} = +0.0007122$.

5 Resource Accounting & Artifact Inventory

- **Pauli Word Universes:** LiH (42,460 terms), BeH_2 (21,930 terms), HF (29,866 terms), H_2O (78,858 terms).
- **Total Wall-Clock Runtime:** ~ 59 seconds (adaptive A-CASE only); ~ 80 minutes including complete SD subspace eigensolves for all 4 molecules.
- **Artifact Inventory** (`molecular_results/`): FCIDUMP files (`lih.fcidump`, `beh2.fcidump`, `hf.fcidump`, `h2o.fcidump`), per-molecule JSON records, and master summary `results_summary.json`.

6 Conclusion

The end-to-end simulation demonstrates that `clifford.qc`'s A-CASE operator subspace solver and sector-restricted backends successfully ingest electronic structure FCIDUMP records, producing exact sector ground references, compact Rayleigh-Ritz eigensolutions, correlation observables, and dynamic Lehmann response spectra without materializing dense $2^n \times 2^n$ statevector matrices. **Chemical accuracy (≤ 1.5936 mHa) is achieved for all four benchmark molecules** when the subspace basis is expanded to the complete set of single-and-double excitations, with errors ranging from machine precision (HF) to +0.7526 mHa (BeH₂).